

Hamza Javed

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Portfolio / hamza1959.github.io/Portfolio

PROFILE

AI Developer with a background in Data Science with Artificial Intelligence, Engineering Management (Climate Change), and Agricultural Sciences. Currently working at Digisoft (NASTP) on climate analytics and smart agriculture solutions, including carbon accounting systems and predictive models.

Experience includes building machine learning pipelines, developing Python-based backends, and integrating models using FastAPI. Worked on real-world applications involving environmental data, sustainability reporting, and industrial analytics.

Able to apply data-driven techniques to solve practical problems, improve system performance, and support decision-making.

EDUCATION

Master of Engineering Management (Climate Change Management) *Aug 2024 – Dec 2025*
NED University of Engineering and Technology — Karachi, Pakistan

Postgraduate Diploma in Data Science with Artificial Intelligence *May 2024 – Dec 2025*
NED Academy — Karachi, Pakistan

- Performed data preprocessing, model development, and evaluation for predictive analytics.

Bachelor of Science in Agricultural Sciences *Oct 2018 – Jul 2022*
Pir Mehr Ali Shah Arid Agriculture University — Rawalpindi, Pakistan

- Developed strong foundation in soil science, crop production, and sustainable agriculture.

PROFESSIONAL EXPERIENCE

AI Developer *Dec 2025 – Present*
Digisoft Transformation Solutions
(National Aerospace Science and Technology Park - NASTP) — Karachi, Pakistan

- Developed an end-to-end Carbon Calculator for emissions estimation to support corporate sustainability reporting and environmental compliance.
- Engineered backend of a Smart Agriculture platform with machine learning modules for crop prediction, fertilizer recommendation, yield estimation, and rainfall forecasting.
- Designed and deployed scalable APIs using FastAPI for real-time model inference.

Data Analyst *Jun 2025 – Nov 2025*
Asani.io
(National Aerospace Science and Technology Park - NASTP) — Karachi, Pakistan

- Performed data analysis and generated performance reports to support business decision-making.
- Developed and deployed machine learning models for forecasting and trend analysis.

Trainee *Apr 2023 – Jun 2023*
Pakistan Council of Scientific and Industrial Research (PCSIR) — Karachi, Pakistan

- Conducted water quality testing, heavy metal detection, and laboratory-based analysis.

Intern *Apr 2022 – Jul 2022*
Barani Agricultural Research Institute — Rawalpindi, Pakistan

- Performed soil and water analysis for agricultural research and field studies

Intern *Jul 2021 – Oct 2021*
Institute of Soil and Environmental Sciences — Rawalpindi, Pakistan

- Conducted comprehensive soil analysis covering NPK, micronutrients, organic matter, pH, and EC to support research activities.

PROJECTS

Smart Agriculture Prediction System

- Developed machine learning models for crop prediction, fertilizer recommendation, yield estimation, and rainfall forecasting.

Carbon Emission Calculator

- Developed a full-stack SaaS Carbon Emission Calculator with React.js (frontend) and Python-based backend, enabling emissions estimation for corporate sustainability reporting

AI Data Assistant (Database-Integrated Chatbot)

- Developed an AI-powered chatbot integrated with database systems to retrieve and deliver structured data through natural language queries.

SKILLS

■ Programming & Development

Python (FastAPI, Flask, Django), REST API Development, React.js (CSS)

■ Machine Learning & AI

Machine Learning, Deep Learning, Predictive Modeling and Gen AI

■ Data Engineering & Processing

Data Preprocessing, Feature Engineering, Model Evaluation

■ Data Analysis & Visualization

Pandas, NumPy, Matplotlib, Seaborn, Power BI

■ Databases & Systems

Database Design, Structured Data Management

■ Domain Expertise

Climate Analytics, Smart Agriculture Systems, Soil & Water Analysis

■ Tools & Other Skills

ArcGIS, Hydrological Modelling, Excel

CERTIFICATIONS

- Python Programming Language
- Business Intelligence
- Data Visualization
- Machine Learning
- On-page SEO
- Freelancing
- Deep Learning

COURSES

Python Programming Language

Sep 2024

- Covered data analysis, visualization, and Python fundamentals

LANGUAGES

- English
- Urdu